36118 Applied NLP · Assessment 2A · Spring 2026

Skills Map

Closing the gap between your degree and the job ad

We put subject outlines, job ads and resumes into one skills vocabulary, so a student can see exactly which skills a role needs that they lack, and which subject fills each gap.

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1 The problem

Subject outlines speak academic. Job ads speak industry. Nothing maps one to the other.

> *"Implement and evaluate advanced machine learning techniques to real-world problems."*
>
> **UTS Handbook 2026, 36120, learning objective 1**

> *"In-depth knowledge of deep learning, Generative AI, and other advanced AI technologies … Experience in building and deploying AI models and solutions."*
>
> **Robert Walters ad, Data Scientist – Python, SQL, Sydney CBD, 23 Sep 2026**

Same skill, different words. Only the ad asks for generative AI, a phrase ESCO has no concept for yet.

2 Three users, one question each

Elective chooser	Which subject moves me closest to my target role?
Job seeker	What does my resume lack for this ad?
Career switcher	How far is my profile from a new role?

Upload a resume or tick subjects taken · pick role and city · gap ranked by demand · electives ranked by gap closed · chat over real ads

3 Six sources, one vocabulary

3,204 Seek Australia Data Scientist ads, 2019–22	**13,960** ESCO v1.2.1 skill concepts	**8,753** O*NET 31.0 tool names, the tooling ESCO lacks
70 UTS Handbook subjects, six courses	**1,355** LinkedIn US data roles, 2023–24	**2,484** Resumes, the student profile

Every count read from the downloaded file, not a description page.

4 Pipeline

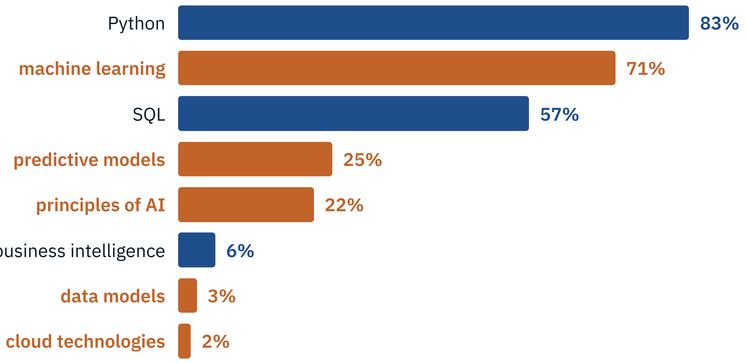
copper stages compare a traditional and a newer method, scored in 8

Input	Clean	Extract skills	Normalise	Match	Output
resume + ad	lowercase, strip	ESCO dictionary → NER, LLM	to ESCO labels	TF-IDF → embeddings	gap, subjects, chat

5 One profile against one role

14 skills the ad demands · **3** resume has · **11** missing

A real Data Scientist ad, Sydney CBD, against a synthetic early-career resume. The biggest gaps are modelling, not tooling.



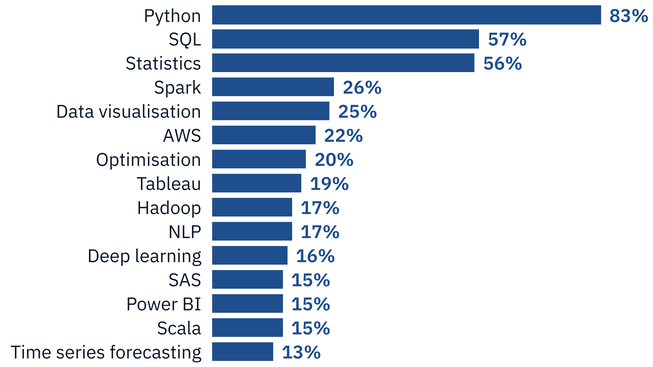
Bar = share of 3,204 Seek Australia ads mentioning the skill. ESCO dictionary extraction, TF-IDF matching. Six required skills are ESCO phrasings no ad uses: counted, not drawn.

6 Which UTS subject fills each gap

machine learning 71% **predictive models** 25%	**36120 Advanced Machine Learning Application** — implement and evaluate advanced machine learning techniques
principles of AI 22%	**36121 Artificial Intelligence Principles and Applications** — general problem-solving approaches in AI; deep neural networks
data models 3%	**94693 Big Data Engineering** — data modelling, distributed processing, streaming
cloud technologies 2%	**42904 Cloud Computing and Software as a Service** — articulate the need for cloud computing

Keyword match over the learning objectives of 70 UTS subjects, six courses; Part B uses embeddings. Percentages are demand from block 7. 42904 is a Master of AI subject, open to MDSI as a free elective.

7 What Australian employers ask for



After the top three, no tool passes 26%. Off the chart: communication appears in 28% of ads, more than Tableau, AWS or Hadoop.

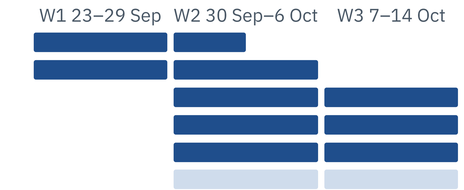
Seek Australia Data Scientist ads 2019–22, n = 3,204, technical skills only. The ads predate 2023: LLMs sit at 0% here and at 15.8% in 247 US Data Scientist ads, 2023–24.

8 How we score each stage

Stage	Baseline → modern	Metric, test set
Extraction	ESCO dictionary → spaCy NER → GPT-4o-mini	P, R, F1; 200 hand-labelled ads
Matching	TF-IDF cosine → sentence embeddings	P@5, MRR; 50 CV–JD pairs
Chat	Keyword search → FAISS RAG + GPT-4o-mini	LLM judge 1–5; 30 questions

Exact ESCO matching covers only 36% of our 133-skill dictionary; 55%, including the whole LLM stack, have no ESCO concept.

9 Three weeks to 14 October



7 Oct pipeline end-to-end · **11 Oct** app and evaluation done · **14 Oct** Part B due

Setup · GitHub AndyPham2004/ANLP-AT2-Skills-Map · Python 3.12 · spaCy, scikit-learn, sentence-transformers, FAISS · Streamlit · OpenAI API, GPT-4o-mini

Ethics · Resumes stay in memory, never on disk; names, emails and phones are masked before any LLM call. Ads used under dataset licences. Results are suggestions, not career advice.

References · European Commission (2025), ESCO v1.2.1 · PromptCloud, Seek Australia job ads 2019–22, Kaggle · Kon (2024), LinkedIn Job Postings 2023–24, Kaggle · Bhawal (2021), Resume Dataset, Kaggle · O*NET 31.0 (2026) · UTS Handbook 2026 · Reimers & Gurevych (2019), Sentence-BERT · Lewis et al. (2020), RAG